Cloud Removal Failure-Mode Analysis: When Inpainting Aids Downstream Land-Cover Classification

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***Abstract*—Optical satellite imagery processing pipelines are persistently constrained by thick, opaque cloud occlusions. While generative image inpainting algorithms optimize low-level visual reconstruction metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS), their utility for downstream semantic tasks remains contentious. This study conducts a pre-specified empirical failure-mode and metric validity analysis to determine whether inpainting cloud-corrupted pixels benefits automated land-use and land-cover (LULC) classification over leaving clouds unaltered, zero-masking missing regions, or co-adapting the downstream classifier. We evaluate on the EuroSAT benchmark (4,050 test tiles per seed, three seeds, ten balanced classes) and on real cloud pairs from the RICE dataset. Three findings emerge. First, a ResNet-18 classifier trained on infilled imagery (fill-aware) outperforms a mask-aware classifier in every seed and every coverage bin, with gains of +5.2, +8.3 and +13.2 percentage points at 10–30%, 30–60% and 60–80% occlusion. Second, classical Fast Marching inpainting (Telea) performs comparably to a deep convolutional U-Net (0.830 vs. 0.826 accuracy at 60–80% coverage), which suggests that lightweight interpolation is adequate once the classifier adapts to the fill distribution. Third, standard perceptual metrics show weak rank correlation with harmful prediction flips (individual fill methods $|\rho| \leq 0.24$; pooled $|\rho| \leq 0.22$). Manual audits of harmful flips suggest that mean-fill infills frequently shift errors into water classes. On semi-transparent clouds, an opacity sweep with the unadapted classifier places a crossover near $\alpha \approx 0.67$, below which, for this classifier, raw cloudy tiles score higher than the restored tiles.**

***Index Terms*—Satellite image inpainting, land-cover classification, cloud removal, failure-mode analysis, metric validity, remote sensing, convolutional neural networks.**

I. Introduction

Optical Earth observation platforms, such as the European Space Agency's Sentinel-2 constellation, continuously monitor terrestrial dynamics for environmental tracking, agricultural yield forecasting, and urban planning [1]. However, persistent atmospheric cloud cover occludes substantial portions of the Earth's surface, creating severe spatial discontinuities in automated processing pipelines. Standard deep learning architectures require fixed-dimension, contiguous raster inputs, forcing practitioners to adopt one of three strategies: discard partially occluded scenes, mask out corrupted regions using zero-padding, or synthesize replacement pixels via spatial inpainting or physical dehazing. Learned approaches to Sentinel-2 cloud removal, including SAR-assisted fusion, have been proposed as well [2].

Within computer vision, image restoration benchmarks evaluate inpainters by measuring pixel-level reconstruction fidelity using Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) [3], and Learned Perceptual Image Patch Similarity (LPIPS) [4]. This evaluation paradigm assumes that higher perceptual and pixel-level fidelity directly yields superior downstream task performance. In operational remote sensing, however, downstream consumers are automated feature extractors and classifiers rather than human observers. Fabricated high-frequency textures produced by neural inpainters can satisfy perceptual losses while introducing synthetic artifacts that cross semantic decision boundaries, and this can trigger misclassifications that are hard to notice.

Moreover, some prior evaluations confound the restoration method with the classifier. Infilled images are scored by classifiers trained only on pristine, cloud-free imagery. When such a classifier is applied to infilled images under heavy occlusion, its performance drops sharply, which can create the impression that inpainting is inherently worse than zero-masking.

To address these issues, this paper conducts a controlled, pre-specified empirical investigation into the utilitarian efficacy of cloud inpainting for satellite land-cover classification. Our research addresses three core questions:

1) **Utilitarian Value:** Does filling cloud-occluded pixels improve classification accuracy over leaving cloudy pixels as-is or zero-masking them?
2) **Classifier Co-Adaptation:** Does adapting the downstream classifier to the artifact distribution of inpainters resolve the degradation observed under heavy cloud cover?
3) **Metric Validity:** Do standard image restoration metrics (PSNR, SSIM, LPIPS) reliably predict or detect downstream semantic classification failure?

Our experimental investigation yields the following contributions:

- **Masking reversal:** The apparent superiority of zero-masking under heavy cloud cover appears to be largely an artifact of evaluating unadapted classifiers. Once the classifier is trained on infilled distributions, inpainting beats masking in every coverage bin and every seed (+13.2 points at 60–80% occlusion).
- **Classical vs. learned inpainting:** Parameter-free Telea interpolation performs comparably to a deep convolutional U-Net for downstream classification (83.0% vs. 82.6% accuracy at 60–80% coverage).
- **Metric validity:** Using tile-level bootstrap confidence intervals over 4,050 test tiles, PSNR, SSIM and LPIPS show weak rank associations with downstream failure, which suggests caution in using them alone as proxies for classification utility.
- **Failure taxonomy:** Manual labelling of 60 harmful flips shows that mean fill mostly produces colour-shift errors, and a 200-tile analysis suggests these errors move terrestrial samples into water classes (*SeaLake*, *River*).
- **Opacity crossover:** For the unadapted classifier on semi-transparent clouds, leaving the cloud in place scores higher than inpainting below $\alpha \approx 0.67$.

II. Related Work

A. Classical and Deep Image Inpainting

Image inpainting techniques fall into two broad classes: partial differential equation (PDE) based interpolation and deep generative modeling. Classical algorithms, such as the Fast Marching Method pioneered by Telea [5] and Navier–Stokes fluid continuity models [6], [7], propagate structural edges from the unoccluded boundary into the missing region using local gradient fields. While computationally trivial and parameter-free, they lack global semantic awareness.

Deep learning approaches formulate inpainting as conditional generative modeling. Convolutional encoder–decoder architectures, such as the U-Net [8], partial convolutions [9], and generative adversarial networks (GANs) [10], synthesize missing pixels by conditioning on high-level receptive fields. In remote sensing, deep paired restorers have been trained on paired multi-temporal datasets such as RICE [11]. These models require heavy GPU training budgets and optimize predominantly for pixel-wise L_1 or perceptual losses.

B. Atmospheric Dehazing

For semi-transparent clouds and atmospheric haze, physical scattering models are widely adopted. The Dark Channel Prior (DCP), introduced by He et al. [12], leverages the statistical rarity of low-intensity pixels in non-sky color channels to estimate atmospheric light and haze transmission. Although effective for outdoor terrestrial photography, its validity on nadir satellite multispectral imagery may be compromised by high ground reflectance and heterogeneous surface albedo.

C. Task-Driven Restoration and Research Gap

Restoration guided by a high-level vision task has been studied for denoising and recognition [13]. Its systematic evaluation in Earth observation remains scarce. Many satellite cloud removal benchmarks stop at computing PSNR and SSIM against ground-truth clear acquisitions [11]. To our knowledge, few studies have investigated co-adaptation, that is, retraining the downstream feature extractor on infilled artifacts. This leaves practitioners with conflicting guidelines on whether to drop, mask, or reconstruct cloud-covered tiles.

III. Methodology

A. Problem Formulation and Scope

Let $\mathbf{I}_{\text{clean}} \in \mathbb{R}^{H\times W\times C}$ denote a pristine C-channel optical satellite tile of spatial dimensions $H \times W$. Cloud occlusion is represented by a binary mask $\mathbf{M} \in \{0,1\}^{H\times W}$, where $\mathbf{M}_{i,j} = 1$ denotes cloud presence. The corrupted image $\mathbf{I}_{\text{cloudy}}$ is modeled via alpha blending:

$$\mathbf{I}_{\text{cloudy}} = (1 - \alpha\mathbf{M}) \odot \mathbf{I}_{\text{clean}} + \alpha\mathbf{M} \odot \mathbf{I}_{\text{cloud}}, \tag{1}$$

where $\mathbf{I}_{\text{cloud}} \in \mathbb{R}^{H\times W\times C}$ represents cloud reflectance, $\alpha \in [0,1]$ represents optical thickness, and $\odot$ denotes element-wise multiplication. For opaque (thick) clouds, $\alpha = 1$, eliminating ground transmission.

An inpainting algorithm $\mathcal{F}$ estimates the missing pixels:

$$\mathbf{I}_{\text{fill}} = (1 - \mathbf{M}) \odot \mathbf{I}_{\text{cloudy}} + \mathbf{M} \odot \mathcal{F}(\mathbf{I}_{\text{cloudy}}, \mathbf{M}). \tag{2}$$

The downstream task is defined by a classifier $\mathcal{C} : \mathbb{R}^{H\times W\times C} \to \mathcal{Y}$, which predicts the categorical land-cover label $y \in \mathcal{Y}$. Our objective is to evaluate the classification risk:

$$R(\mathcal{C}, \mathcal{F}) = \mathbb{E}_{(\mathbf{I},y)} \left[\mathbb{I}\left(\mathcal{C}(\mathbf{I}_{\text{fill}}) \neq y\right)\right]. \tag{3}$$

Scope restriction: Inpainted pixels are synthetic estimates rather than real ground observations. We therefore evaluate land-cover classification across complete raster tiles and exclude safety-critical uses such as military reconnaissance, disaster evacuation and meteorology.

B. Dataset Profiles and Pre-Specification

We evaluate on two complementary benchmarks:

1) **EuroSAT:** 27,000 georeferenced Sentinel-2 RGB tiles (64×64 pixels, 10 m ground resolution) across 10 balanced land-cover classes [1]. A stratified split (seed 42) gives 18,900 training tiles (70%), 4,050 validation tiles (15%) and 4,050 full test tiles (15%).
2) **RICE:** A real-world remote sensing benchmark of paired clear and cloudy images without categorical labels [11]. We use RICE-I (626 held-out test pairs with thin, semi-transparent clouds) and RICE-II (717 held-out test pairs with thick, opaque cloud masses).

All evaluation protocols, bin boundaries, statistical tests and decision rules were pre-specified in `ANALYSIS_PLAN.md` before the primary multi-seed runs.

C. Synthetic Cloud Protocol and Leakage Prevention

Synthetic cloud masks are generated across three coverage intervals: light (10–30%), moderate (30–60%), and heavy (60–80%). Cloud textures are synthesized by sampling multi-scale Gaussian-filtered noise octaves:

$$\mathbf{N}_{\text{comb}} = 0.6\,\mathbf{G}_{\sigma_1} * \mathcal{N}_1 + 0.3\,\mathbf{G}_{\sigma_2} * \mathcal{N}_2 + 0.1\,\mathbf{G}_{\sigma_3} * \mathcal{N}_3, \quad (4)$$

where $\sigma_1 \in [6, 12]$, $\sigma_2 \in [3, 6]$, and $\sigma_3 \in [1, 2.5]$, followed by thresholding at the required target coverage percentile. The semi-transparent condition uses a fixed $\alpha = 0.5$.

Leakage prevention: Training masks for classifier adaptation are drawn from an independent procedural shape-mask family (random ellipses and rectangles), while test masks are generated exclusively with the multi-scale fractal cloud simulator. The test classifier therefore never encounters test cloud geometries during training.

D. Evaluated Inpainting Arms

We benchmark four restorative techniques alongside raw baselines:

1) **Cloudy as-is:** the corrupted image $\mathbf{I}_{\text{cloudy}}$ evaluated directly.
2) **Zero-masking (black fill):** occluded pixels set to zero, $\mathbf{I}_{\text{mask}} = (1 - \mathbf{M}) \odot \mathbf{I}_{\text{clean}}$.
3) **Mean fill:** occluded pixels set to the mean channel reflectance of the visible context.
4) **Nearest neighbour (NN):** Euclidean distance transform propagation, assigning each missing pixel the value of the closest valid pixel.
5) **Telea inpainting:** Fast Marching edge propagation [5].
6) **Scratch U-Net:** a 4-channel (RGB $+\mathbf{M}$) convolutional U-Net trained from scratch on EuroSAT for 30 epochs with Adam (learning rate 10^{-3}, batch size 64) and a weighted loss that penalizes in-cloud error (Eq. 5).
7) **Dehazing (DCP):** the Dark Channel Prior [12], applied globally and restricted to the cloud mask.

The U-Net loss is

$$\mathcal{L}_{\text{unet}} = 5\,\|\mathbf{M} \odot (\hat{\mathbf{I}} - \mathbf{I})\|_1 + \|(1 - \mathbf{M}) \odot (\hat{\mathbf{I}} - \mathbf{I})\|_1. \quad (5)$$

E. Downstream Classifier Training Regimes

All classifiers use an ImageNet-initialized ResNet-18 backbone [14], fine-tuned end-to-end for 3 epochs with Adam (learning rate 3×10^{-4}, batch size 128) and cross-entropy loss, checkpointed at the minimum validation loss epoch. We use three training regimes:

- **Clean classifier ($\mathcal{C}_{\text{clean}}$):** trained solely on pristine, cloud-free EuroSAT tiles, representing traditional production pipelines.
- **Mask-aware classifier ($\mathcal{C}_{\text{mask}}$):** trained on cloud-free tiles corrupted with black geometric shape masks (80% probability), forcing the network to recognize incomplete spatial patterns.
- **Fill-aware classifier ($\mathcal{C}_{\text{fill}}$):** trained on tiles restored by all four fill methods (Telea, U-Net, NN, mean) alongside 20% clean tiles, adapting the feature representations to inpainting artifacts.

Experiments are replicated across three independent random seeds (0, 1, 2). Each seed retrains all classifiers and regenerates the fill-aware training set. On clean test tiles the clean classifier reaches 95.7–96.7% accuracy across seeds.

F. Failure-Mode and Metric Validity Framework

To separate inpainting-induced errors from baseline classifier mistakes, we define a **harmful flip** as a test instance where the classifier predicts the correct label on the pristine image but misclassifies the infilled image:

$$\text{Harmful} = [\mathcal{C}(\mathbf{I}_{\text{clean}}) = y] \wedge [\mathcal{C}(\mathbf{I}_{\text{fill}}) \neq y]\,. \quad (6)$$

Conversely, a **helpful flip** occurs when a misclassified cloudy image is restored to the correct category:

$$\text{Helpful} = [\mathcal{C}(\mathbf{I}_{\text{cloudy}}) \neq y] \wedge [\mathcal{C}(\mathbf{I}_{\text{fill}}) = y]\,. \quad (7)$$

To test whether standard reconstruction metrics predict downstream failure, we compute the Spearman rank correlation coefficient (ρ) between harmful flips and full-tile SSIM, full-tile LPIPS, in-mask PSNR, and in-mask SSIM. Resampling uses tile-level bootstrapping ($B = 1{,}000$ iterations) over all 4,050 test instances in each of 12 experimental conditions to obtain 95% confidence intervals.

G. Real-Cloud Evaluation Protocol

RICE has no class labels, so accuracy cannot be computed. We report restoration fidelity using LPIPS (lower is better) and **classifier agreement** (computed with the fill-aware classifier): the fraction of tiles on which the classifier predicts the same label for the restored tile as for the true clear reference. Agreement is a prediction-consistency proxy rather than proof of semantic correctness. Following the analysis plan, tiles whose clear-image prediction is River or SeaLake, and tiles whose prediction is unstable under small perturbations, are excluded from the agreement computation. This leaves 428 of 626 RICE-1 tiles and 249 of 717 RICE-2 tiles.

IV. Experimental Results

A. EuroSAT Land-Cover Classification Performance

Table I presents downstream classification accuracy across three independent seeds on the full 4,050-tile EuroSAT test split.

The main findings are as follows.

1) **Fill-aware vs. mask-aware:** The pre-specified decision rule was to retrain on filled data only if the fill-aware classifier beats the mask-aware classifier in every seed and every bin. This holds: for each of the four fill methods, the fill-aware accuracy exceeds the mask-aware baseline in all nine seed–bin cells. The Telea gains are $+5.2$, $+8.3$ and $+13.2$ percentage points at 10–30%, 30–60% and 60–80% occlusion. On a 200-tile subset, paired bootstrap intervals for Telea minus the mask-aware baseline are $+5.0$ [0.5, 9.5], $+8.5$ [3.5, 14.5] and

TABLE I
DOWNSTREAM LAND-COVER ACCURACY ON EUROSAT (MEAN ± SD OVER 3 SEEDS, $N = 4{,}050$ TEST TILES PER SEED).

Restoration strategy	10–30%	30–60%	60–80%
Fill-aware classifier (C_{fill}), adapted to fills			
Mean fill	0.934 ± 0.009	0.885 ± 0.010	0.789 ± 0.008
Nearest neighbour	0.943 ± 0.009	0.903 ± 0.012	0.799 ± 0.010
Telea inpaint	$\mathbf{0.950 \pm 0.007}$	$\mathbf{0.916 \pm 0.008}$	$\mathbf{0.830 \pm 0.015}$
Scratch U-Net	0.948 ± 0.005	0.914 ± 0.006	0.826 ± 0.007
Cloudy as-is	0.682 ± 0.060	0.348 ± 0.103	0.151 ± 0.056
Mask-aware classifier (C_{mask}), fair mask baseline			
Masked (black fill)	0.898 ± 0.014	0.833 ± 0.017	0.698 ± 0.022
Clean classifier (C_{clean}), unadapted baseline			
Cloudy as-is	0.730 ± 0.041	0.343 ± 0.032	0.137 ± 0.014
Nearest neighbour	0.946 ± 0.009	0.810 ± 0.021	0.441 ± 0.039
Telea inpaint	0.942 ± 0.006	0.716 ± 0.032	0.267 ± 0.004
Scratch U-Net	0.939 ± 0.007	0.689 ± 0.039	0.255 ± 0.019
Mean fill	0.912 ± 0.018	0.533 ± 0.074	0.144 ± 0.006

TABLE II
OPACITY SWEEP (30–60% COVERAGE, $N = 500$ TEST TILES, CLEAN CLASSIFIER; DCP APPLIED TO THE WHOLE TILE; 95% CI $\approx \pm 0.04$).

α	Cloudy	Telea	Mean	DCP (whole)
0.20	**0.826**	0.584	0.478	0.584
0.35	**0.712**	0.586	0.478	0.496
0.50	**0.650**	0.584	0.478	0.406
0.65	0.592	0.584	0.478	0.346
0.80	0.520	**0.584**	0.478	0.302
1.00	0.408	**0.584**	0.478	0.226

+15.5 [9.0, 22.0] points. These subset intervals are wider because the sample is smaller.

2) **Resolution of the masking reversal:** In unadapted clean classification, zero-masking appeared superior to inpainting under heavy clouds (0.698 vs. 0.267 for Telea). This appears to be an artifact of domain shift. Once the classifier is co-adapted to infilled features, inpainting rises from 0.267 to 0.830, clearly exceeding zero-masking at heavy cover.
3) **Equivalence of classical and deep inpainting:** Telea and the scratch U-Net perform comparably in every bin (0.950 vs. 0.948, 0.916 vs. 0.914, 0.830 vs. 0.826). Once the classifier is co-adapted, fast PDE interpolation gives similar downstream accuracy to the deep model.
4) **Nearest neighbour as a baseline:** Nearest neighbour stays within about 3 points of Telea and U-Net, while mean fill has the lowest accuracy among the inpainters.

B. Cloud Opacity Sweep and Dehazing Analysis

Table II reports accuracy across opacity levels $\alpha \in [0.20, 1.00]$ on 500 test tiles at 30–60% coverage, using the clean classifier. This sweep is a separate evaluation from Table I. Its clean-classifier Telea accuracy (0.584) is lower than the main-split value (0.716). We report both as measured and do not pool them.

Telea and mean fill are constant across α because inpainting overwrites every pixel inside the mask, discarding the underlying reflectance. Semi-transparent clouds likely retain some high-frequency ground signal, so cloudy accuracy falls more slowly as opacity drops. Interpolating between the 0.65 and 0.80 rows, the curves cross at about $\alpha \approx 0.67$. The 0.65 point (0.592 vs. 0.584) already lies inside the interval, so we treat the crossover as an approximate threshold rather than a sharp bound. This result holds for the clean classifier only. We did not run the fill-aware classifier across opacity levels, so we make no claim for it.

Dehazing with the Dark Channel Prior does not appear to help. Restricted to the mask, DCP reaches 0.758 at $\alpha = 0.20$, which still falls short of raw cloudy imagery (0.826). Applied to the whole tile, as in the table, DCP falls to 0.226 under opaque clouds.

TABLE III
REAL-CLOUD RESTORATION ON RICE (LPIPS WITH 95% BOOTSTRAP INTERVAL; AGREEMENT ON THE STABLE SUBSET).

Restoration method	LPIPS ↓	Agreement (%)
RICE-1 (thin clouds; $n = 428$ of 626 tiles)		
Cloudy as-is	$\mathbf{0.149}$ $[\mathbf{0.135}, \mathbf{0.165}]$	58.9
Paired restorer (big)	0.174 [0.149, 0.205]	**71.3**
Paired restorer (small)	0.178 [0.154, 0.211]	69.6
Dark channel prior	0.189 [0.158, 0.231]	69.6
Telea inpaint	0.329 [0.303, 0.358]	50.5
Nearest neighbour	0.278 [0.256, 0.304]	47.7
Mean fill	0.355 [0.331, 0.382]	55.1
RICE-2 (thick clouds; $n = 249$ of 717 tiles)		
Cloudy as-is	0.402 [0.364, 0.436]	46.2
EuroSAT U-Net	0.374 [0.355, 0.391]	58.6
RICE U-Net (small)	$\mathbf{0.318}$ $[\mathbf{0.295}, \mathbf{0.338}]$	65.5
Telea inpaint	0.324 [0.298, 0.346]	65.1
Nearest neighbour	0.341 [0.313, 0.367]	56.6
Mean fill	0.323 [0.299, 0.344]	81.5
Paired restorer (big)	0.336 [0.309, 0.358]	**84.3**
Dark channel prior	0.544 [0.512, 0.576]	45.8

C. Evaluation on Real Clouds: RICE Benchmark

Because real clouds have no categorical labels, we assess restoration fidelity with LPIPS and prediction consistency with classifier agreement, computed on the stable subset described in Section III-G.

Findings: On thin real clouds (RICE-1), leaving the cloud as-is gives the lowest perceptual error (LPIPS = 0.149). Paired restorers raise classifier agreement to about 70%, but their LPIPS intervals overlap the cloudy baseline. On thick clouds (RICE-2), the RICE-trained U-Net has lower LPIPS (0.318) than the EuroSAT-trained U-Net (0.374), although the two models differ in more than training data. The paired restorers give the highest PSNR on both datasets and, on RICE-1, the highest SSIM. On RICE-2, however, their LPIPS is worse than Telea, mean fill and the RICE U-Net. On RICE-2, mean fill (81.5%) and the paired restorer (84.3%) show similar agreement with overlapping intervals. Because the stable subset drops water-predicted tiles, high agreement does not rule out water-class collapse on real tiles. A manual review of real tiles would be needed to test that.

D. Metric Validity Analysis

Table IV evaluates the predictive validity of standard image quality metrics for harmful flips of the fill-aware classifier across 4,050 test tiles in 12 experimental conditions.

TABLE IV
SPEARMAN RANK CORRELATION (ρ) BETWEEN QUALITY METRICS AND HARMFUL FLIPS (POOLED, $N = 4{,}050$ TILES, 95% BOOTSTRAP CI).

Metric	Expected sign	Pooled ρ	95% CI
Whole-tile SSIM	Negative	-0.21	$[-0.22, -0.20]$
Full-tile LPIPS	Positive	$+0.22$	$[+0.21, +0.23]$
In-mask PSNR	Negative	-0.12	$[-0.14, -0.11]$
In-mask SSIM	Negative	-0.09	$[-0.11, -0.08]$

TABLE V
WITHIN-METHOD SPEARMAN RANK CORRELATIONS (ρ) WITH HARMFUL FLIPS.

Fill method	SSIM (full)	LPIPS	PSNR (mask)	SSIM (mask)
Mean fill	-0.21	$+0.22$	-0.13	-0.06
Nearest neighbour	-0.23	$+0.24$	-0.13	-0.13
Telea inpaint	-0.20	$+0.21$	-0.11	-0.09
Scratch U-Net	-0.19	$+0.21$	-0.11	-0.09

TABLE VI
HARMFUL CLASSIFICATION FLIP RATES BY FILL METHOD (%, FILL-AWARE CLASSIFIER, 4,050 TILES).

Infill method	10–30%	30–60%	60–80%
Telea inpaint	**1.5**	**4.9**	**13.9**
Scratch U-Net	1.7	5.4	14.1
Nearest neighbour	2.1	6.0	15.9
Mean fill	3.2	7.5	17.9

Metric failure: All metrics correlate in the expected direction, and each interval excludes zero, but the associations are weak. The strongest metric, LPIPS ($\rho = +0.22$), shows only a weak association with harmful flips. Within individual coverage bins the correlations weaken further: whole-tile SSIM ranges from -0.08 to -0.12, full-tile LPIPS from $+0.07$ to $+0.12$, and in-mask PSNR from -0.06 to -0.14.

To check that pooling across fill methods does not drive this result, we computed rank correlations within each method (Table V).

The within-method coefficients are close to the pooled values. High PSNR or low LPIPS therefore does not reliably guarantee that an infilled tile keeps its semantic label.

E. Harmful Flip Rates and Empirical Failure Taxonomy

Table VI gives harmful-flip rates by fill method for the fill-aware classifier.

Telea has the lowest harmful-flip rate in every bin, consistent with its top accuracy. To find out why the methods fail, we labelled 60 verified harmful flips by hand (15 per method) into four failure modes:

1) **Colour shift:** 13 of 15 mean-fill flips. Flat mean reflectance removes spatial gradients, and the resulting smooth, dark regions are often predicted as *SeaLake* or *River*.
2) **Texture hallucination:** 10 of 15 nearest-neighbour flips. Copied boundary pixels create repeated textures that look like *AnnualCrop* or *PermanentCrop*.
3) **Lost geometric structure:** 8 of 15 Telea flips and 9 of 15 U-Net flips. Diffusion and convolution smooth over linear infrastructure, severing narrow features such as *Highway* and *River*.
4) **Contextual boundary artifacts:** the remaining cases, mainly spectral bleeding at urban–vegetation edges (*Industrial*/*Residential* confusion).

The water-class pattern appears broader than the manual labels alone show. On a 200-tile subset scored with the unadapted classifier, 91% of mean-fill harmful flips land in *SeaLake* or *River*. The same holds for the U-Net (85%) and Telea (77%), and for NN (45%). The U-Net is not a flat fill, so water collapse does not seem specific to flat infills. We therefore describe it as a sink for errors at heavy coverage that is strongest for mean fill. Helpful flips are rare by comparison: in the 200-tile analysis, no method recovers more than 3% of tiles in any bin.

TABLE VII
TRANSFER-LEARNING INPAINTER ACCURACY SCORED BY THE FILL-AWARE CLASSIFIER (SEED 0, EUROSAT).

Inpainting backbone	10–30%	30–60%	60–80%
Scratch U-Net (ref.)	**0.945**	**0.914**	**0.832**
ResNet-18 (ImageNet)	0.894	0.842	0.712
ResNet-18 (satellite)	0.895	0.834	0.674
ResNet-18 (random)	0.819	0.766	0.626

F. Transfer Learning and the Spatial Bottleneck

We benchmarked U-Nets with a ResNet-18 encoder under three initializations against the scratch U-Net. Identical decoders, loss (Eq. 5) and 30-epoch schedules were used. The satellite-pretrained variant received 10 epochs of self-supervised block reconstruction on unlabelled EuroSAT tiles. Table VII reports accuracy scored by the fill-aware classifier for seed 0 only.

ImageNet initialization improves on random initialization, but every ResNet-18 variant stays below the scratch U-Net: ImageNet by 5–12 points and random initialization by 13–21 points. We suspect the cause is the 2×2 spatial bottleneck at `layer4` for 64×64 inputs, which may discard boundary detail that inpainting relies on. We did not test this directly. The table reports one seed only, so these gaps are indicative rather than established.

V. CONCLUSION AND FUTURE WORK

This investigation clarified the main trade-offs in satellite cloud inpainting for land-cover classification. Our primary conclusions are:

1) **Fill and co-adapt:** Under heavy occlusion (60–80%), filling missing pixels and training the classifier on infilled imagery beats the mask-aware baseline by $+13.2$ points,

and the gain holds in every seed and bin. This meets the pre-specified decision rule.

2) **Simple inpainters are adequate:** Parameter-free Telea interpolation performs comparably to a deep U-Net for downstream accuracy in this setting, which avoids expensive GPU training.
3) **Metrics are weak proxies:** PSNR, SSIM and LPIPS show only weak rank associations with harmful classification flips.
4) **Opacity threshold:** For the unadapted classifier, leaving semi-transparent cloud in place beats inpainting below $\alpha \approx 0.67$. We have not measured this for the fill-aware classifier.

Limitations: EuroSAT clouds are synthetic noise on 64×64 RGB tiles. They lack cirrus geometry, topographic shadows and infrared bands. RICE evaluation uses classifier agreement as a proxy because no categorical labels exist, and the stable subset excludes many tiles. Transfer-learning results come from a single seed. The opacity sweep is a separate 500-tile evaluation whose clean Telea value differs from the main split, and the 200-tile analyses are used only to support the paired intervals.

Future work: Future efforts will extend co-adaptation to multispectral Sentinel-2 bands (red-edge, NIR, SWIR), evaluate full-scene acquisitions ($> 256 \times 256$ pixels), run the fill-aware classifier across opacity levels, and explore end-to-end architectures that jointly optimize inpainting and classification.

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APPENDIX: AUTHOR CONTRIBUTIONS (CREDIT)

Author contributions follow the Contributor Roles Taxonomy (CRediT).

- **P. Sai Dharanidhar Ram (24AI10004):**
  - *Conceptualization:* Co-formulated the research questions, pre-specified decision rules, and evaluation scope.
  - *Methodology:* Designed the three classifier training regimes (clean, mask-aware, fill-aware) and the four inpainting evaluation arms.
  - *Software:* Developed the primary evaluation harness, classical baseline scripts, downstream ResNet-18 classifier implementations, and the RICE paired-restorer and Dark Channel Prior baselines.
  - *Formal analysis:* Computed multi-seed statistical significance, binomial intervals, and paired bootstrap accuracy differences.
  - *Writing – original draft:* Authored the Methodology, Experimental Results (EuroSAT classification and transfer learning), and Abstract.
- **Ruthvik Devarakonda (24AI10023):**
  - *Conceptualization:* Co-developed the hypothesis on classifier co-adaptation and the failure taxonomy framework.
  - *Data curation:* Audited the EuroSAT dataset, established stratified reproducible splits, and processed the real-cloud RICE benchmark.
  - *Software:* Implemented the procedural cloud and shape-mask simulator, metric evaluation module, scratch U-Net, and real-cloud evaluation scripts.
  - *Validation and investigation:* Executed the 60-sample manual failure audit, verified prediction shifts into water bodies, and performed the tile-level bootstrap Spearman rank correlations.
  - *Writing – review and editing:* Authored the Introduction, Related Work, Metric Validity Analysis, Failure Taxonomy, and Conclusions.
  - *Project administration:* Managed reproducible environment configurations, repository checkpoints, and compliance with course guidelines.